

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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A Project Phase 2 Report on

**IOT BASED SOLAR POWERED INVERTER AND SMART
IRRIGATION SYSTEM**

A Project work submitted in partial fulfilment for the award of the degree

**Bachelor of Engineering in Electronics and
Communication Engineering**

Submitted by

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**Under the Guidance of
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2025-2026**



**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
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Certified that the Major Project Phase 2 entitled **“IOT Based Solar Powered Inverter and Smart Irrigation System”** is carried out by **Pooja (1AY22EC065), Radhika (1AY22EC071), Shreya T N (1AY22EC094) and Suma M V (1AY22EC101)** in the partial fulfillment for the award of the degree of Bachelor of Engineering in Electronics and Communication Engineering of Visvesvaraya Technological University, Belagavi during the year **2025-2026**. It is certified that all corrections/suggestions indicated for the assessment have been incorporated in the report deposited in the departmental library. The Major Project Phase 2 Report has been approved as it satisfies the academic requirement in respect of Major Project Phase 2 (BEC786) prescribed for the Bachelor of Engineering Degree.

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DECLARATION

We the students of seventh semester Electronics and Communication Engineering, Acharya Institute of Technology, Soladevanhalli Bengaluru -560107 declare that work entitled “IOT Based Solar Powered Inverter And Smart Irrigation System” has been successfully completed under the guidance of Dr Ragunath B H, Department of ECE, Acharya Institute of Technology, Bengaluru. This dissertation work is submitted to Visvesvaraya Technological University in partial fulfilment of the requirements for the award of Degree of Bachelor of Engineering in Electronics and Communication Engineering during the academic year 2025-2026. Further the matter embodied in the Major Project Phase 2 report has not been submitted previously by anybody for the award of any degree or diploma to any university.

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ACKNOWLEDGEMENT

The satisfaction and euphoria that accompany the successful completion of a task would be incomplete without the mention of the people who made it possible and without their constant guidance and encouragement success would not have been possible.

We are grateful to the institute **Acharya Institute of Technology** for its ideas and inspiration for having provided us with the good infrastructure, laboratory, facilities, and inspiring staff which has made this Project Phase 2 successfully.

We would like to express my sincere gratitude to **Dr. C. K. Marigowda** Principal, AIT for all the facilities that has been extended by him throughout my work.

We heartily thank and express my sincere gratitude to **Dr. Rajeswari**, HOD, Dept. of ECE AIT for her valuable support and a constant source of enthusiastic inspiration to steer us forward.

We would like to express my sincere gratitude to the Internal Guide **Dr Raghunath B H**, Assistant Professor, Dept. of ECE, AIT for his invaluable guidance and support.

We would like to express my sincere gratitude to the Project Coordinators **Dr. Jayalaxmi H** Associate Professor and **Dr. Nikita Kar Chowdhury**, Assistant Professor, Dept. of ECE, AIT for their valuable guidance and support.

Finally, We would like to express my sincere gratitude to my parents, all teaching and non-teaching faculty members, and friends for their moral support, encouragement, and help throughout the completion of the Project.

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ABSTRACT

Water is a critical input into agriculture in nearly all aspects having a determining effect on the eventual yield. Good seeds and fertilizers fail to achieve their full potential if plants are not optimally watered. In spite of water scarcity farmers also face electrical crisis due to increased demand for electrical energy with the growing population. solar powered auto irrigation system gives the solution for the above problem by using the soil moisture sensor based on solar power. Solar energy is best way for the irrigation purpose to overcome energy crisis problem. The solar panel will extract energy from the sun and convert into electrical energy which is stored in the battery. Automatic irrigation using solar power can be efficiently used for the proper management of irrigation. Proper irrigation increases fertility rate of field and can get maximum production of crops by increasing yields and the quality of the crop by improved management of water during critical plant growth stages. The benefit of solar based automatic irrigation using soil moisture is the field gets continuous water and plants grows fast. The proper management of water helps to get rid of problem from water scarcity. If automated system is not working then pump can be also operated manually. The manual operation can be done through the manual switching system. Another benefit of the manual operation is we can also use the pump to supply the water. The soil moisture sensor measures the soil resistivity or volumetric moisture contain of soil in terms of voltage. An automated irrigation system was developed for optimized usage of water in agriculture. The system has a distributed wireless network of soil moisture, humidity and temperature sensors placed at the root zone of plants. Arduino microcontroller gets the sensor information, triggers the output devices. A mechanism was developed depending on the threshold values of the soil moisture, humidity and temperature sensor and type of plant. Programmed into Arduino to control the level of water fed to the irrigation field. Sensors and Arduino are connected with each other with wire. Because of low energy consumption and cost, the system has potential to be useful in water-limited geographically isolated areas.

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CHAPTER 1

1.1 INRODUCTION

Country agriculture has a very important role in the development of the country. Although most of the land around the globe is cultivated land, much of that land is being under utilized because of a lack of sufficient information about how to grow crops or due to a lack of water. Because many of these lands are dependent upon monsoon rains to water them, when the seasons are unfavorable the crops are either late growing or drying up. Additionally, since rainfall can be sporadic, farmers need to purchase additional water to irrigate their fields and this increases their costs of production. As we all know there is often a lack of a viable market for agricultural products and therefore, investing large sums of money into the production of those products is not economically feasible. However, even though there may be adequate amounts of water available in certain areas, poor management of the water leads to crop failure. Furthermore, excessive flooding will also lead to decreased rates of production of crops and/or decreased fertility of the land. Solar based, automatic irrigation utilizing soil moisture sensors, will assist in providing an appropriate amount of water to the crops at regular intervals and will help to alleviate the problems associated with water scarcity and excessive consumption of water through an automatic system. Many different technologies have developed numerous methods and types of irrigation systems to address the issue of fulfilling water requirements; however, as the demands for fossil fuels continue to rise globally and as fossil fuels are considered non-renewable resources, it would be beneficial to develop irrigation systems that utilize renewable energy sources. The utilization of renewable energy, such as solar energy, is one method of addressing the issues of both energy crisis and load shedding. Therefore, the use of solar based irrigation systems are well suited to meet the needs of irrigation. This type of irrigation utilizes automatic irrigation techniques that are controlled by a soil moisture sensor placed in the field. The soil moisture sensor measures the volumetric moisture content or soil resistivity found within the soil. When the moisture content in the soil is low or the soil is dry, the pump will operate to provide the necessary water. The control mechanism of this system is provided by a microcontroller and therefore, the microcontroller is the central component of this irrigation system.

In the event that the automatic system fails, the pump can still be started by means of a manual switch. The manual switch does not shut off the irrigation system; rather, the farmer must be present at the time of use. This system combines solar energy, IoT communication and sensor-

based automation to provide efficient irrigation and real-time monitoring for advanced farming practices. At the heart of the system is an ESP32 microprocessor, which continuously monitors environmental factors, including soil moisture, temperature, humidity, rainfall and light, utilizing various sensors. The ESP32 microprocessor determines whether or not to irrigate the crops based upon soil moisture levels and automatically starts the irrigation pump only when necessary. In addition, if the system detects rainfall, it immediately stops the pump to avoid wasting water. Furthermore, to increase the sustainability of the system, it utilizes solar power with an inverter and therefore continues to operate without interruption regardless of whether or not there is grid power available to the field. Additionally, when light levels are low, a Light-Dependent Resistor (LDR) automatically adjusts the lighting to these low levels, and an infrared (IR) sensor detects the presence of animals within the field, causing a buzzer to sound and the electric fence to activate to protect the crops from animal damage. Real-time data monitoring and control via a Telegram-based interface are made possible by the use of IoT via Wi-Fi, allowing the farmer to receive immediate alerts and system status updates directly to their smartphone. On-site, a Liquid Crystal Display (LCD) displays sensor readings such as temperature, humidity, moisture level and system status. This project demonstrates how an effective combination of renewable energy, automation and IoT technologies can be used to create smart irrigation, better crop protection, less farmer involvement and better resource management for modern, precision agriculture and smart farming systems.

1.2 PROBLEM DEFINITION

Traditionally irrigated systems have been subject to the effects of unreliable electrical power, overuse of water, and a lack of automation in their use of water. The result has been an increase in labor required for irrigation, wasted water, and decreased crop yields. Traditionally, farmers are also affected by changes in the weather and animals accessing the field, which both negatively impact agricultural production.

The proposed smart irrigation system is based on Internet of Things (IoT) technology and utilizes a combination of solar power, sensor inputs, and an ESP32 microcontroller to provide automation for irrigation, and remote alert capabilities for real-time monitoring of water usage and other system functions.

1.3 OBJECTIVES

- The primary focus of this project, in terms of addressing the overall goal of developing a single, integrated system using both solar energy and IoT technology to provide a highly efficient automated irrigation system for the agricultural field, is to resolve two major challenges currently faced by the agricultural community; i.e., the challenge of being dependent on the use of energy, and the challenge of inefficiently utilizing water.
- By using IoT technology to monitor and control the irrigation process in real time, the environmental conditions of the field are continuously monitored through a network of environmental sensors (soil moisture, temperature and humidity sensors). The data from these sensors are then read by the system and utilized by the microprocessor or embedded computer to determine how often, and the amount of water required, to be delivered through the irrigation system to minimize the unnecessary use of water while preventing loss.

CHAPTER 2

2.1 LITERATURE REVIEW

1. “Solar Powered Smart Irrigation Using IoT”

Authors: Kuruvalli Vinay Kumar Reddy, Challa Raviteja, Venkateswar Reddy Rashmi Priyadarshini B. K. (2021)

Description: This work focuses on an IoT-based irrigation setup powered by solar energy, which automatically switches the irrigation pump on/off based on soil moisture. The use of valves and microcontrollers eliminates the need for manual intervention

2. “Smart Irrigation System Using Arduino with Solar Power”

Authors: Saleh Elkelani Babaa, Muneer Ahmed, Babatunde Samuel Ogunleye, Shahid Ali Khan, Salim Ahmed Al-Jahdhami, John Regan Pillai (2020)

Description: This system integrates solar power with IoT sensors controlled by Arduino to automate irrigation. The design demonstrates energy harvesting through PV panels and automation of water distribution in agriculture for enhanced sustainability

3. “IoT Based Automated Irrigation System Using Solar Power”

Authors: Shaikh Zameer S. Shabbir, Vaishali J. Kothale, Dnyaneshwar S. Wayal, Apeksha S. Sarkate, Sagar R. Dukare, Mangesh J. Nemade (2020)

Description: Presents an IoT-driven automatic irrigation system using solar energy. The system uses soil moisture, temperature, and humidity sensors to optimize water delivery with cloud-connected monitoring via mobile app, thus reducing manual intervention.

4. “IoT based Irrigation Powered by Solar Voltaic System”

Authors: Abdullah Bin Suroor, Pratik Singh, Harsh Anand, Dr. J. P. Kesari (2020)

Description: This paper details a solar photovoltaic powered irrigation system that uses IoT sensors and controllers to monitor soil conditions and automate irrigation, highlighting a low-cost sustainable approach suited for rural agriculture.

5. “Simulation and Construction of a Solar Powered Smart Irrigation System Using IoT”

Authors: Samson Dauda Yusuf, Salam-Layode Dupe Comfort, Ibrahim Umar, Abdulmumini Zubairu Loko (2022)

Description: The study combines simulation and practical construction of a solar-powered smart irrigation system with Node MCU and mobile app integration (Blynk), enabling real-time remote monitoring and efficient irrigation control.

6. “Solar Powered Smart Irrigation System”

Authors: B. Ravikumar, V. Supriya, E. Meenaloshini, S. U. Teja (2024)

Description: Designed to automate irrigation using IoT sensors and solar panels. The system also incorporates moisture and level sensors with remote control capability to optimize water usage and reduce wastage

7. “Based Smart Solar Inverter for Solar Power Generation”

Authors: P. Prakash, S. Nandha Kumar, K. Sathish Kumar, S. Sreejith (2023)

Description: Focuses on an IoT-enabled solar inverter system that monitors panel output, battery status, and power distribution in real time. The smart inverter uses communication modules to optimize energy generation and storage

8. “IOT Based Self-Tracking Solar Powered Smart Irrigation System”

Authors: Abdul Raheman, M. Kameswara Rao, B. Vamsi Reddy, T. Ravi Kumar (2018)

Description: This study presents a solar powered smart irrigation system that tracks soil moisture and water availability. The system uses sensors and ATmega8 microcontroller to automate irrigation based on soil moisture thresholds, helping farmers manage water effectively

Summary Of Literature Review

The literature review shows that significant research has been carried out in the areas of IoT-based smart irrigation systems and solar-powered energy solutions, highlighting their importance in achieving sustainable agriculture and efficient energy management. Most researchers agree that combining IoT technology with renewable energy sources such as solar power helps reduce dependence on conventional electricity and enables system deployment in remote and rural areas. Several studies focus on solar-powered smart irrigation systems, where soil moisture, temperature, and humidity sensors are used to automate irrigation. These systems supply water only when required, thereby reducing water wastage and improving crop productivity. Researchers have demonstrated that IoT platforms allow farmers to monitor field conditions remotely through mobile or web applications, reducing manual effort and improving decision-making. However, many of these systems primarily emphasize irrigation control and do not deeply analyze power management or inverter performance. Other works concentrate on IoT-enabled solar inverter systems, which monitor solar panel output, battery status, and load conditions in real time. These systems improve reliability, enable early fault detection, and enhance energy utilization. Despite these advantages, most inverter-based studies operate as standalone energy-monitoring solutions and are not tightly integrated with agricultural automation.

CHAPTER 3

3.1 REQUIREMENTS SPECIFICATION

➤ **Hardware Requirements:**

- ESP32 Controller
- Water Pump
- Soil Moisture Sensor
- LCD
- Power Supply
- Relays
- Humidity And Temperature Sensor
- Buzzer
- Battery

➤ **Software Requirements:**

- Arduino Suite
- Embedded C

3.1 ESP32:

ESP32 is an SoC (System on Chip) and has been one of the increasingly popular chips in the past few years. Whether ESP32 became popular because of the increasing IoT (Internet of Things) industry applications or whether IoT industry application increased due to the emergence of ESP32 is something that could be subject to debate and discussions among experts and enthusiasts. You might know 10 people who were involved in designing firmware for some IoT application, and there are chances that 7 to 8 of those 10 people might have dealt with ESP32 at least once in their past projects. Now comes the question of what this excitement and hype are for ESP32? In what way was it able to achieve such widespread popularity in such a short duration? Let's learn something and then proceed to understand the reason behind ESP32's increasing popularity. Here are the specifications of ESP32 WROOM 32 Module: ESP32 has many additional features and specs when it comes to ESP8266 and it would become cumbersome to write specs for this ESP32 Tutorial guide for everyone to implement and understand easily for first time users of ESP32 like me as well. Therefore, I have jotted down some of the key specs of ESP32 below: But for that, I would request you to refer to the datasheet

for the full set of specs for thorough understanding and analysis of ESP32's technicalities and implementations.

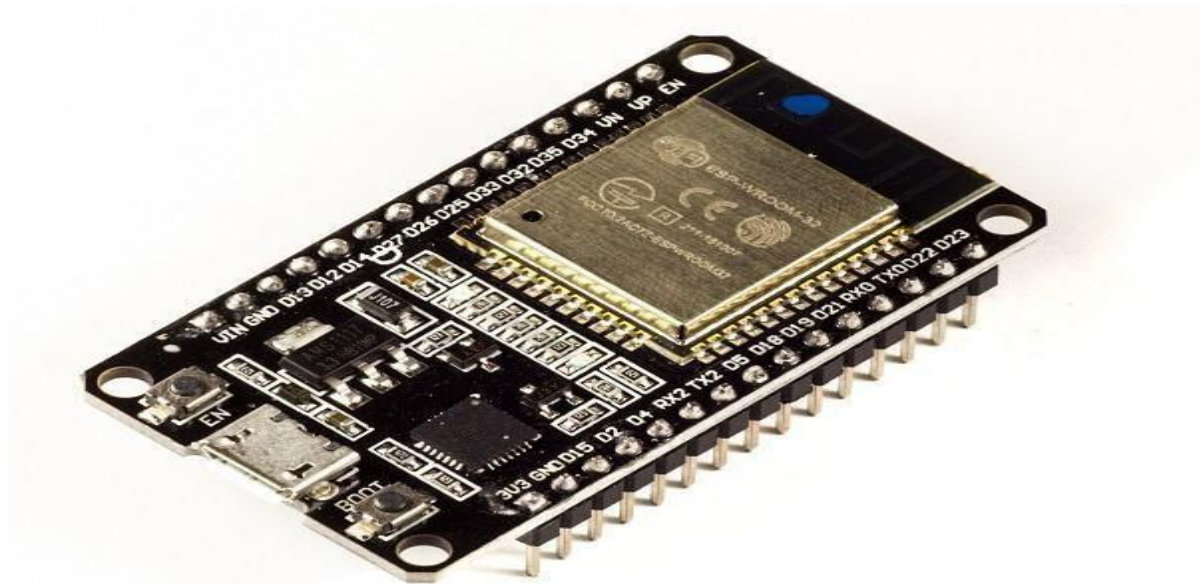


Fig 3.1 : ESP32

- Integrated Crystal– 40 MHz
- Module Interfaces- SPI, UART, I2C, PWM, DAC, GPIO, ADC, pulse counter, capacitive touch sensor
- Integrated SPI flash– 4 MB
- ROM– 448 KB (Booting & Basic Operations)
- SRAM– 520 KB
- Connected Devices Bluetooth, BLE, Protocols- WiF
- On–chip sensor– Hall sensor
- Operating temperature range– –40 – 85 degrees Celsius
- Operating Voltage– 3.3V
- Operating Current– 80 mA (average)
- These specifications provide a clear picture as to why ESP32 is so popular. An IoT device will require that its μ C meet several conditions. From the last chapter, we know that sensing, processing, storage and transmission are the primary functional components of any IoT device. Therefore, the first thing the μ C has to do is connect to various types of sensors. The μ C must be able to interface with all common

communication protocols necessary for connecting to sensors; UART, I2C and SPI, etc. Also, the μ C should have ADC capability and pulse counting capability. In addition to the capabilities listed above, the μ C should be capable of interfacing with capacitive touch sensors. This means most common types of sensors can be connected directly to an ESP32.

- Secondly, the μ C should have the capacity to perform element processing for the input data coming from the sensors at times with high speed and should have enough memory to retain the data. The ESP32 has a max operating frequency of 40 MHz This is very fast. Also, this has two processors. This allows parallel processing. This is an additional advantage. Finally, a total SRAM capacity of 520 KB is good enough for processing a very large amount of data. Any number of Popular processing algorithms like FFT processing, peak detection, RMS value computation, and so on can easily occur in ESP32. Also, for storage purposes, ESP32 has moved a step ahead compared to normal microcontrollers. This has a file system in Flash. Out of the total storage capacity of 4MB of Flash in ESP32, a total of 1.5 MB is used for SPIFFS (SPI Flash File System) storage. Think of a mini-SD Card residing inside the microchip itself. Not only data storage is possible. Storing Text files, Images, HTML/CSS Files, and a whole lot more inside SPIFFS is possible. People have shown very beautiful Web Pages running on a Wi-Fi Server created using ESP32. The data used for display has been stored in the HTML file stored inside SPIFFS
- For data transmission, ESP32 comes with a built-in WiFi and Bluetooth stack which has been a game-changer in its own right. No need for any additional connection module (GSM module and LTE module) for testing your cloud communications. All you need is your ESP32 board and your WiFi up and running, and you are good to go! ESP32 also enables you to use your WiFi module in Access Point as well as Station Modes! While ESP32 supports TCP/IP, HTTP, MQTT, and other standard communication protocols, it also supports HTTPS! Yes, you read that correct! It comes with a crypto-core or a crypto-accelerator which actually is a piece of hardware that has a specific job associated with it: accelerate encryption! So you not only get to communicate with your web server, you get to do it all over a secure connection! BLE functionality is also a highly valuable aspect for many projects and applications. Of course, you could communicate via LTE, GSM, and LoRa modules with ESP32 as well! So in terms of 'data transmission as well', ESP32 breaks all molds

- It would be charging through the nose considering all the specs that ESP32 is packed with. This is the most amazing thing about ESP32 dev modules that the price at which they are available is somewhere around the range of ₹500. This is not all—the size of the actual chip is also small (25 mm × 18 mm, including the antenna section).

Specification of ESP32

- Single or Dual-Core 32-bit LX6 Microprocessor with a maximum CPU Speed of 240 MHz
- 520 KB of SRAM, 448 KB of ROM, and 16 KB of RTC SRAM.
- Ethernet MAC for Physical LAN Communication (requires PHY).
- Comes with 802.11b/g/n Wi-Fi technology supporting 150 Mbps data transfer rate
- Support for Classic Bluetooth Specification v4.2 and BLE.
- 34 Programmable GPIO
- Serial Connectivity: includes 4 x SPI, 2 x I2C, 2 x I2S, 3 x UART
- 1 Host controller for SD/SDIO/MMC and 1 Slave controller for SDIO/SPI
- Up to 18 channels of 12-bit SAR ADC and 2 channels of 8-bit DAC
- Motor PWM and up to 16 channels of LED PWM.
- Control over Secure Boot.
- Cryptographic Hardware Acceleration for AES, Hash (SHA-2), RSA, ECC and RNG.

VERY IMPORTANT NOTE: ESP-WROOM-32 Modules, there are many ESP32 Boards available in Markets. Each board has a different layout, pinout, and characteristics. I have ESP32 Board with 30 Pins (15 on both sides). But there are some boards available in markets with a different number of Pins, some have 36 Pins, some have slightly less Pins. So, it is very important to identify Pins before doing any actions.

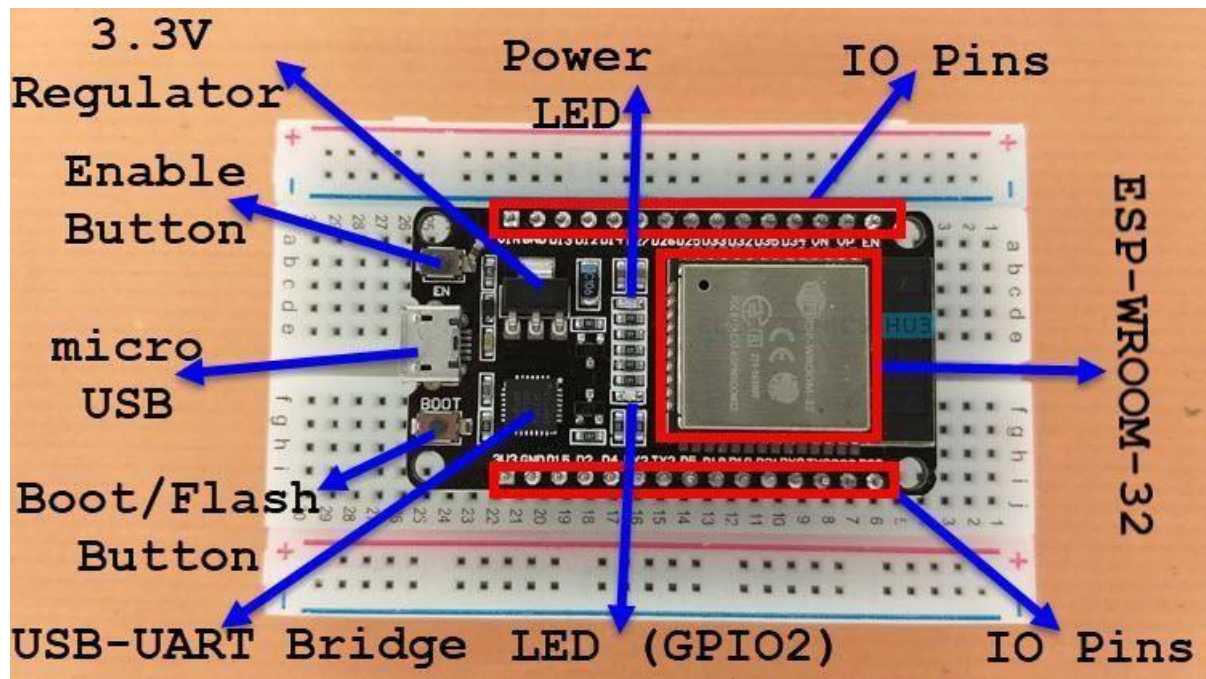


Fig 3.1.1: ESP32 Board

As seen from the image, the ESP32 Board has the following components:

- ESP WROOM-32 Module
- CP2012 USB – UART Bridge IC
- AMS1117
- 3.3V Regulator IC
- Enable Button (for Reset)
- Boot Button (for flashing)
- Power LED (Red)
- Two rows of IO Pins with 15 pins on each side
- User LED (Blue – connected to GPIO2)
- micro–USB Connector (for power and programming)
- Some passive components

One interesting thing about the USB to UART IC is that the DTR and RTS pins are used to put the ESP32 into programming mode automatically (whenever needed) and to reset the board after programming.

Pinout of ESP32 Board

Take a look the pinout diagram of the ESP32 Development Board

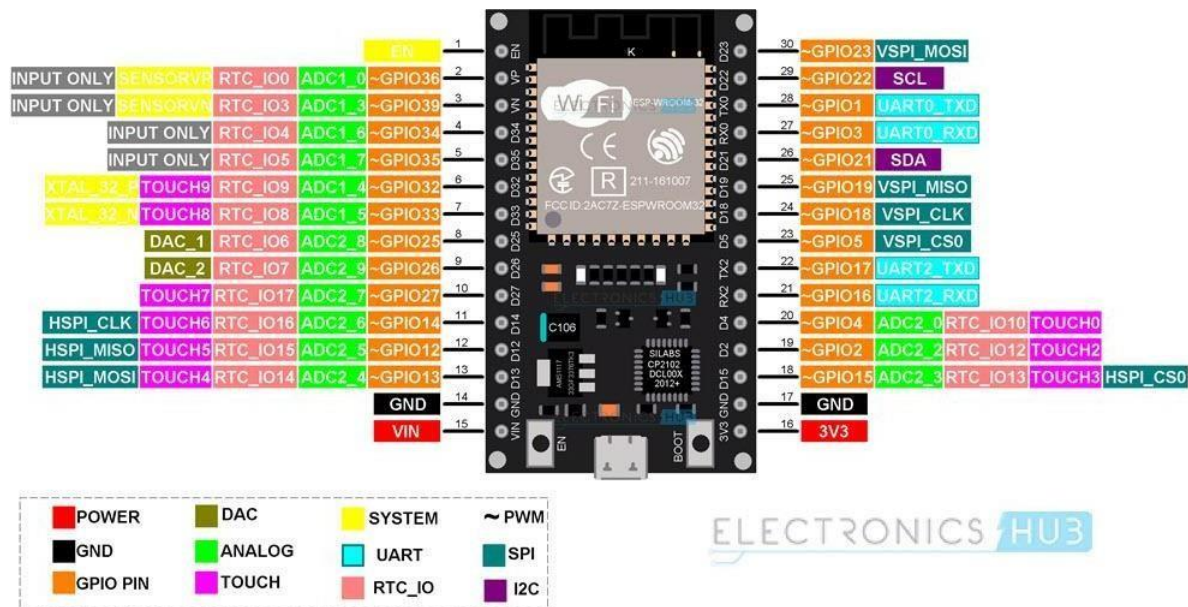


Fig 3.1.2: Pinout of ESP32 Board

This pinout is for the 30 – pin version of the ESP Board. I will explain the pin out of both the 30 – pin as well as the 36 – pin version of the ESP Boards

3.2 LCD display

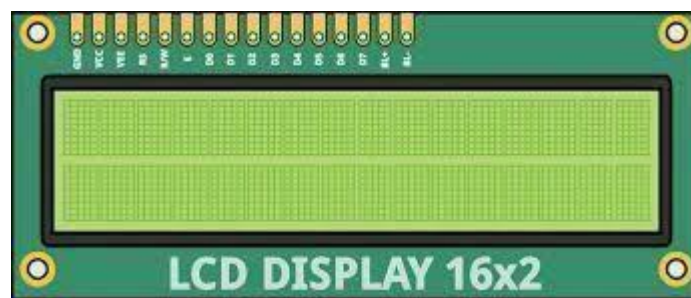


Fig 3.2: LCD display

A liquid-crystal display (LCD) is a flat-panel display LCD is also applied in a vast variety of applications, including computer displays, TVs, instrument panels, airplane instrument panels, and signages; indoors and outdoors. Smaller LCD displays are also frequently utilized in portable consumer electronics, including digital cameras,

watches, calculators, and mobile phones, including smartphones. LCD screens are also applied on consumer electronics products, for example, DVD players, video game consoles, and clocks. LCD screens have become a substitute for the bulkier and heavier cathode ray tube screens in all practical applications. LCD screens are also offered in a vast variety of display sizes compared to plasma and cathode ray tube screens, with LCD screens available for purchase from tiny digital watches to big screens for TVs. As LCD monitors do not use phosphors, they resist image burn-in effects when an image of fixed content is placed on an LCD for an extended period of time (e.g., when an air schedule table frame appears inside an indoor sign). Connecting an LCD to an Arduino The 16x2 LCD has a total of 16 pins. As shown in the table below, are data lines (pins 7-14), eight of the pins two are for power and ground (pins 1 and 16), three are used to control the operation of LCD (pins 4-6), and one is used to adjust the LCD screen brightness (pin 3). The remaining two pins (15 and 16) power the backlight. The details of the LCD terminals are as follows:

Terminal 1	GND
Terminal 2	+5V
Terminal 3	Mid terminal of potentiometer (for brightness control)
Terminal 4	Register Select (RS)
Terminal 5	Read/Write (RW)
Terminal 6	Enable (EN)
Terminal 7	DB0
Terminal 8	DB1
Terminal 9	DB2
Terminal 10	DB3
Terminal 11	DB4
Terminal 12	DB5
Terminal 13	DB6
Terminal 14	DB7
Terminal 15	+4.2-5V
Terminal 16	GND

Table 3.2: Lcd Terminals

3.3 Relay:

A relay is an electrically operated switch. Current flowing through the coil of the relay creates a magnetic field which attracts a lever and changes the switch contacts. The coil current can be on or off so relays have two switch positions and they are double throw (changeover) switches. The relay's switch connections are usually labeled COM(POLE), NC and NO.



Fig 3.3: Relay

In most relay systems, the relay operates through the use of an electromagnet in operating a mechanical switch; however, other systems exist, including solid-state relay systems. Relays are applicable in situations involving control of a circuit by a lower power signal or in situations whereby a number of circuits are commanded by a single signal. The early relay systems were applied in telegraphs used for long-distance communication as amplifier systems; they reused a signal received from one circuit on a different circuit.

3.4 Soil Moisture Sensor:

The two copper probes of the sensor function as the sensing probes. These two copper probes are inserted into the soil specimen whose water contents are being measured. The conductivity of soil varies depending on the amount of water it contains. It increases due to the increase in water contents that form a conductive path between two sensing probes.

FC-28 soil moisture sensor is a simple breakout board that allows you to find out the moisture levels in soil and similar substances. The soil moisture sensor is very easy-to-use stuff. The two big exposed pads on this sensor act as probes for this sensor, which work as a variable resistor when taken as a whole. The more water your pads will be full and hence result in a smaller resistance and a larger "an out."

FEATURES

1. Digital output threshold adjust potentiometer
2. Power and digital output indicator or LED's
3. Analog and digital outputs
4. Mounting hole for easy installation

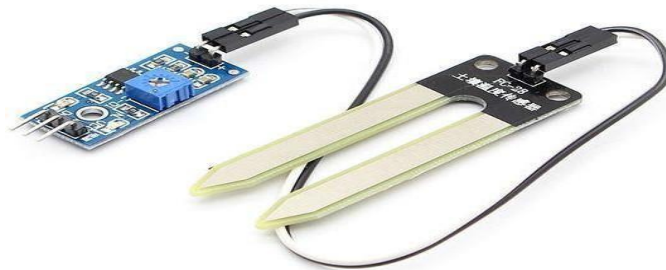


Fig 3.4: Soil Moisture Sensor

3.5 Humidity Sensor (DHT11):

Humidity sensor: It is used to detect the vapours present in the atmosphere. The variation in RH (Relative Humidity) of the surroundings will help to display the readings.

Humidity refers to the amount of water vapor in the air. The quantity of humidity in the air has effects on some physical, chemical, and biological processes. In an industry, humidity affects the business cost of the commodity, the health and safety of the personnel. Thus, in semiconductor industry and control system industry, measurement of humidity has great significance. Humidity measurement relates to measuring the quantity of moisture in gas that can range from a mix of water vapor, nitrogen, or argon to pure gas. Humidity measurement has two types, which depend on the units for measurement. They are a relative humidity sensor and an Absolute humidity sensor. DHT11 is a digital temperature and humidity sensor. DHT11 is a low-cost digital sensor that takes measurements for temperatures and humidities. The sensor can quickly be connected with any micro controller such as Arduino, Raspberry Pi. Thus DHT11 digital sensor for measuring humidity and temperatures can quickly be connected with any micro controller for measuring humidity and temperatures instantly. DHT11 Humidity & Temperature sensor comes in two packages -in the form of a sensor and a module. The two are similar to each other; the main difference exists in a pull-up resistor and a Power ON LED. DHT11 is a relative humidity sensor. The measurement of the surrounding environmental air

with the help of DHT11 sensor includes a thermistor for measurement and a capacitive type humidity sensor for humidity measurement.

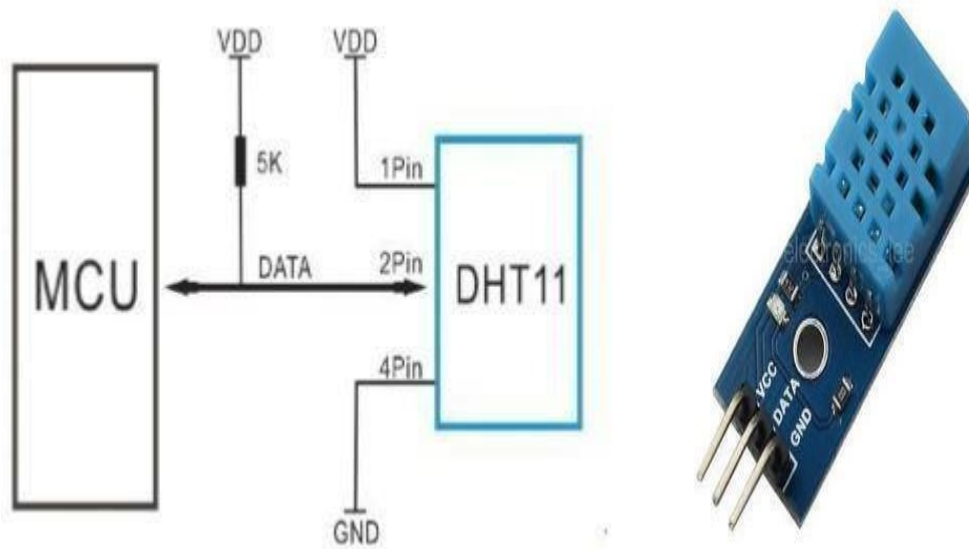


Fig 3.5: Humidity Sensor (DHT11)

3.6 Buzzer:



Fig 3.6: Buzzer

A buzzer is a very compact yet highly efficient device for adding sound functions to our project or system. A buzzer is actually very compact with a 2-pin structure that can be easily interfaced in bread boards, Perf Boards, or even in PCBs that result in it being widely used for most electronic applications.

Buzzer Pin Configuration

Pin Number	Pin Name	Description
1	Positive	Identified by (+) symbol or longer terminal lead. Can be powered by 6V DC
2	Negative	Identified by short terminal lead. Typically connected to the ground of the circuit

Table 3.6: Buzzer Pin Configuration

There will be two buzzers that can be readily available. One of which is shown in this figure and is named Simple Buzzer. When electric power is applied in this buzzer, a reality sound is produced; this goes like this: "Continues Beeeeeeppp." The other one is named "Readymade buzzer," and this will be larger in size compared with the former one and will produce "Beep. Beep. Beep." Sound due to an internal oscillation circuit. But this one shown here will be most useful because this can be altered with some other circuits, and thus this will readily fit in our project. In this, electric power needs to be applied in a DC power supply ranging from 4-9V. A 9V battery can also be applied in this buzzer. In this buzzer, it is always preferable to use +5V or + 6V regulated DC power supply. In this buzzer, this is mostly applied in collaboration with a switching circuit in order to switch this ON/OFF at some time intervals.

3.7 Infrared sensor:

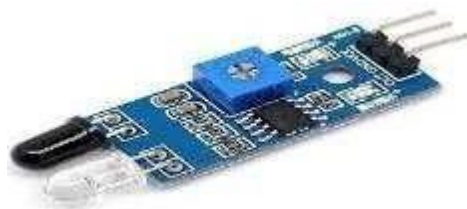


Fig 3.7: Infrared Sensor

It is an electronic device. It is used to detect certain properties within its environment by emitting irradiations of infrared rays. Able to measure the heat and the moving object.

This type of sensor only measures the infrared radiation and does not have the ability to emit the radiation. Description This sensor will only measure the infrared radiation and will not have the ability to infrared radiation and will not have the ability to emit the infrared radiation and is known by the name “Passive IR Sensor.” The objects within the infrared region produce different forms of thermal radiation. This type of sensor will only measure the infrared radiation and will not have the ability to emit the radiation. Description This type of sensor will only measure the infrared radiation and will not have the ability to emit the infrared radiation and is known by the name “Passive IR Sensor.” The objects within the infrared region produce different forms of thermal radiation. It has the ability to measure the heat and also the motion of the object using the IR sensor that will only measure the infrared radiation and will not have the ability to emit the infrared radiation and is known by the name “Passive IR Sensor.”

3.8 Solar panel:



Fig 3.8: Solar panel

Solar cells are classified as photovoltaic cells, whether it is solar or artificial lighting. Apart from functioning as an energy-generating cell, another application of a solar cell is as a photodetector, for example, infrared detectors, which can detect light or any other electromagnetic wave close to the visible spectrum, or measure the intensity of the light.

There are three primary necessities for the functioning of a photovoltaic

- Light absorption, a process by which either an electron-hole pair or an exciton is created
- The separation of charge carriers of opposite types.
- The individual extraction of these carriers into an external circuit.

On the other hand, a solar thermal collector is one which supplies heat through the absorption of sunlight, with the aim of direct heating or the subsequent production of electricity through the utilization of the absorbed heat. The term "photo electrolytic cell" is on the other hand the name of photovoltaic cells, as in that of Edmond Becquerel and the dye-sensitized solar cell, or an electrolytic cell that splits water directly into hydrogen and oxygen solely through the use of solar light. The colloquial term solar panel is that of the photo-voltaic module.

A PV module is a combination of photo voltaic cells structured in a frame work for mounting. Photo voltaic cells work on the source of sunlight and produce direct current electricity. A group of PV modules is termed a PV panel, and a group of panels constitutes an array. The arrays of a photovoltaic cell provide solar electricity to the equipment of an electrical circuit.

3.9 Battery:



Fig 3.9: Battery

A battery is an assembly of one or more electrochemical cells that have connections for providing electric power to devices like flashlights and electric cars. In this regard, for a battery that is providing electric power to an electric circuit, its positive terminal will be the cathode while its negative terminal will be the anode. A negative terminal marked is considered to be the source of electrons that will follow an electric circuit to the positive terminal. In this regard, when the battery is connected to an electric load circuit, there occurs an electron transfer reaction that produces low energy products from high energy substances ranging from thin cells of phones to large lead acid cells and even lithium ion cells in cars, to large battery packs that occupy an entire room to offer standby and emergency powers to

telephone stations and computer centers respectively. The free-energy difference appears as electrical energy in the outer circuit. Batteries vary in shape and size: from the small cells that power hearing aids and watches, through notebook computers battery cells, small and thin cells that power smart phones, larger lead acid batteries or li-ion batteries in autos, and down to enormous battery banks that power phone exchanges and data centers.

3.10 Arduino IDE:

The "Arduino Integrated Development Environment (IDE)" is a cross-platform application for Windows, macOS, and Linux, developed using the "Java" programming language. The IDE can be used for writing programs or uploading them into "Arduino Compatible" boards, but also for other development vendor boards with the assistance of "Third Party Cores." The IDE supports "C" and "C++" languages through some special conventions of code arrangement and structuring. The "Arduino IDE" is one of the official software of "Arduino" and, trust me, compilation of code is so easy that an ordinary person, not having any background or technical knowledge, can also try to get their hands dirty in the learning process of this software." "The main code, also known as a sketch, developed from the IDE platform will finally compile into a "Hex File" which will then be uploaded into the controller of the board."

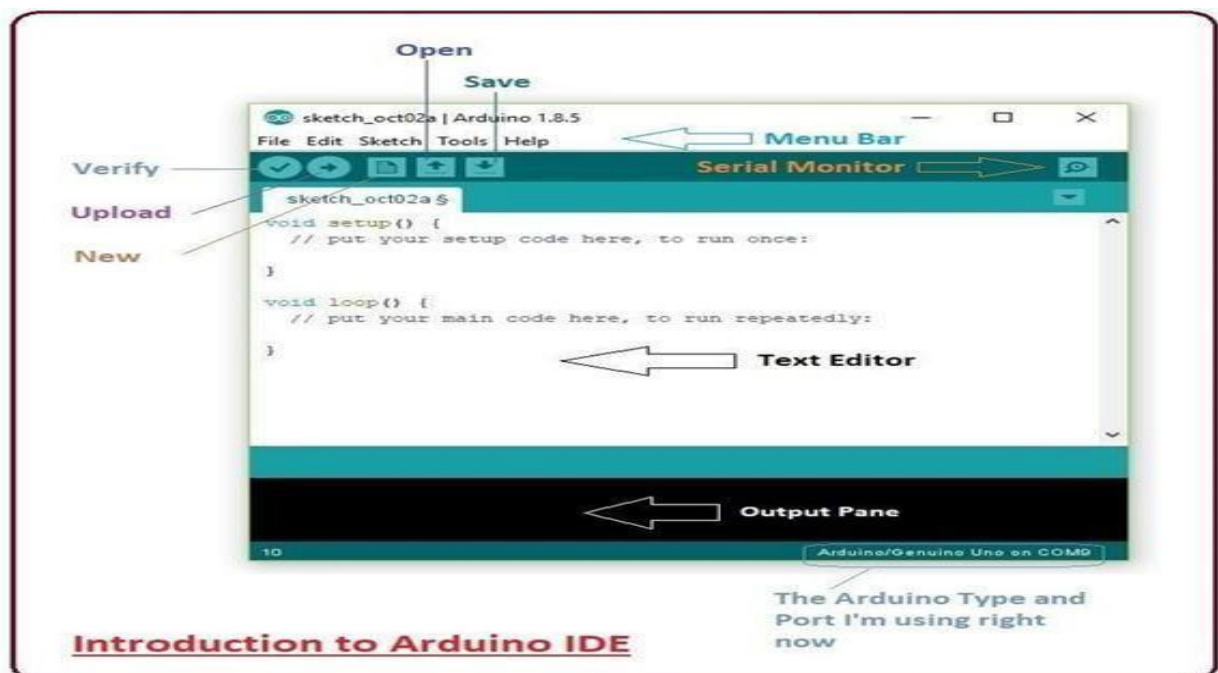


Fig 3.10: Arduino IDE

3.11 Embedded C:

Embedded C is an extension of the C programming language that helps to write efficient programs for an embedded device. Embedded C is not a part of C. C programming is the most widely used programming for an embedded processor/controller. Assembly languages are also used to implement those parts of the program for which very high accuracy of timing is required, and so on.

The Arduino IDE (Integrated development Environment) is fully developed into functionality with full of libraries, and with the purpose of programming the Arduino UNO in Embedded C language only, to develop the project by the Arduino IDE since the IDE can compile code for both Arduino and AVR language.

- The software developed for a smaller embedded system using an 8051 microprocessor will very often consist solely of assembly language code. Many projects will allow this to work effectively, since the amount of code that will have to be developed will not exceed 8 kilobytes, and, in addition, will not be very complicated. When an engineer is responsible for the development of the software as well as the hardware, he or she may very often want to write the software in assembly language code.
- One of the problems that might be experienced in project development using assembly code is that it might be hard to read and maintain. Again, if the code is not commented. Another aspect would be that the code reusable from most assembly language projects would probably be very minimal. This would be significantly improved by developing the project using C code. A C code program would be easier to read compared to the assembly version.
- A C program has a high degree of structure, and thus it is easy to understand and support. Owing to its modularity, it can readily be made applicable for reuse of code from different projects. The breakup of code into functions shall improve its structure and result in functions that can be reapplied or reused in another program in order to save time. A high-order programming language such as C helps and allows a programmer to program in code that closely resembles a human's way of thinking rather than assembly code. A programmer shall devote maximum time to designing algorithms of a system rather than concentratedly working upon its implementation. It shall save considerably on time spent during debugging.
- The use of a language like C means that the programmer does not have to be intimately familiar with the architecture of a particular processor.

CHAPTER 4:

IMPLEMENTATION

4.1 OVERVIEW

Regarding the Implementation of the project, there is a requirement for appropriate system planning. The implementation of automated systems for the agricultural industry has led to a remarkable difference in the effective reduction of costs. The implementation of the automated system has helped in the development of the irrigation industry, as it has led to revolutions in the industry. The project, therefore, has served as the knowledgeable or non-expert system-based observation method for field observation for the purpose of identifying the dryness. Additionally, the final users of the project include the farmers, who are the backbone of our nation. This is based on the reasoning that the system functionality entails automated functionality related to the use of the device and the functionality of the application from the mobile device. Regarding the implementation of the project described for classic designs for the comprehension of the implementation, the initial part is required.

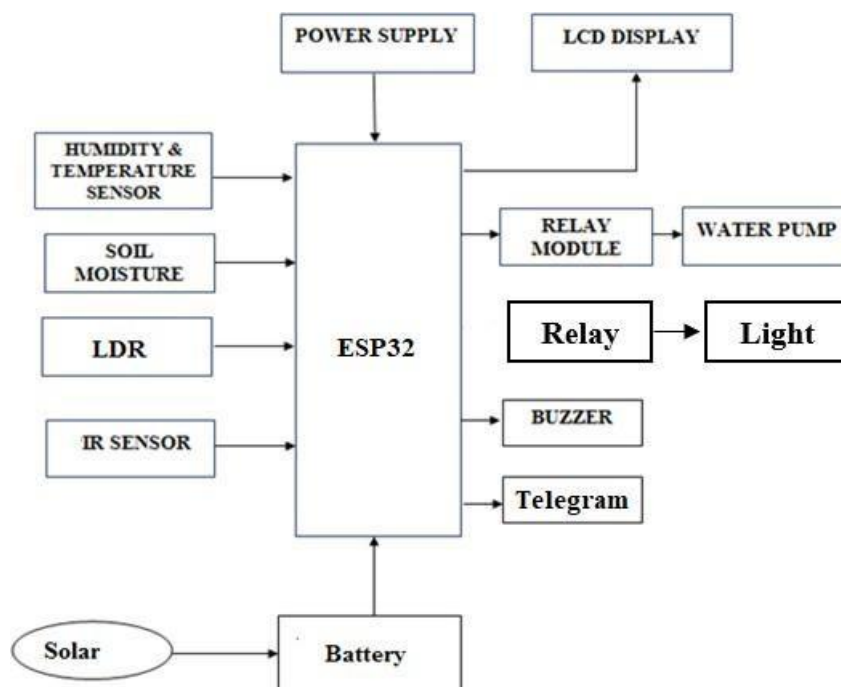


Fig 4.1: Architecture Block Diagram

4.2 METHODOLOGY

System Design: It is required to define the needs and specification for designing the smart irrigation system and design the system by incorporating ESP32, a sensor, a relay, a water pump, and many more features into the system.

Sensor Calibration: The soil moisture sensor, humidity sensor, and LDR sensor should be calibrated for the precise measurement and according to the prevailing environment in the targeted area.

Arduino Programming: Create an Arduino program that can be used to read data from the sensors while making decisions according to predetermined algorithms. Create control circuits for the water pump or relay that can be used to control irrigation using data from the sensors.

Intruder Detection Logic: The logic for identifying intruders should be implemented in the IR sensor. The response to the identification of intruders should include turning on the alarm.

Incorporating the LCD Display: Connecting the LCD Display to the Arduino to display live data regarding the soil moisture and system humidity and related information.

Communication with Telegram: Develop logic for sending messages and alerts to farmers based on system events, such as low battery levels or irrigation completion. Program the ESP32 to establish a Wi-Fi connection and enable communication with the central server or cloud platform

Integration of Solar Cell and Battery: Allow the implementation of a connection between the solar panel and the charge controller and the battery for the charging and discharging processes. Enable the creation of power management logic for effective energy consumption and prevention of the overcharge and discharge cycles of the battery.

Relay and Water Pump Control: Control of the Relay and Water Pump. Connect the relay to the Arduino for controlling the water pump based on the soil moisture level or other factors. Install a safeguard that will not allow more water than the capacity can support, or that will protect the water pump.

Remote Monitoring Setup: Establish a secure and stable connection between ESP32 and the preferred cloud service or server for the purpose of remote monitoring and control. This will lead to the implementation of encryption and authentication processes for securing the data.

User Interface Design: A design for the user interface that will be implemented on the LCD display and ESP32 has been discussed below, so that there shall be no problems for the farmer in reading out the most important information about the system.

Deployment: The deployment of Solar-Based Smart Irrigation System on the agricultural land that has been identified. Farmers will be trained on the use of the aforementioned Solar-Based Smart Irrigation System.

Monitoring and Maintenance: Use a remote monitoring service for monitoring the functionality of the smart irrigation system. Schedule a maintenance service for cleaning, calibration, and upgrading the software.

CHAPTER 5:

5.1 TESTING AND VALIDATION

Testing and validation were carried out to ensure that the IoT Based Solar Powered Inverter and Smart Irrigation System operates accurately, reliably, and efficiently under real-time conditions. This phase verifies that the hardware components, embedded software logic, and IoT communication work together as intended. The objective was to confirm correct automation of irrigation, proper utilization of solar energy, and effective remote monitoring through IoT.

5.2 Testing Power Supply (Solar Panels and Inverters)

Purpose: Confirm that the system receives a stable power supply from the Solar Panels through the inverter for all times.

Test Procedure

- A solar panel was connected to an inverter to power the system.
- Variation of load conditions were achieved by turning on the relay, pump, buzzer, and LCD.
- Continuous power supply during varying levels of light intensity.

Expected Outcome

- A steady and unbroken flow of electricity to ESP32 and peripherals.

Actual Result

- The system functioned without resets or loss of operation.
- The inverter supplied backup power when the amount of solar radiation is low.

5.3 Wi-Fi Connectivity & IoT Communication Testing

Purpose: To confirm that there is a successful connection of the ESP32 via Wi-Fi and the ESP32's Telegram based IoT communication.

Test Procedure

- ESP32 successfully connects to the previously set up Wi-Fi network.

- The startup notification was sent from the ESP32 to the Telegram application.
- /Hello and /Status commands were used to test the functionality of the Telegram application with the ESP32.

Expected Outcome

- A successful Wi-Fi connection.
- Sending real time notifications through Telegram in response to user input or sensor readings

Actual Result

- Notification confirming startup.
- A correct display of sensor data and system status in Telegram.

5.4 Testing of Soil Moisture Sensor for Automatic Irrigation System Operation

Purpose: Validate irrigation system operation that is dependent upon soil moisture.

Test Procedure

- The soil moisture sensor was tested in both low and high moisture (wet) soils.
- Relay control operation of pump was monitored.

Expected Outcome

- When soil moisture level is Low → Pump is ON
- When soil moisture level is Sufficient → Pump is OFF

Actual Result

- Pump Turned ON when operating under a low moisture condition.
- Pump Turned OFF when soil moisture level became sufficient.

5.5 Rain Sensor Testing

Purpose: prevent irrigation during rainfall.

Test Procedure

- The water droplets are placed on the rain sensor.

- Pump and alert display observed.

Expected Outcome

- Rain detection → Pump OFF and notification sent.

Actual Result

- The pump immediately ceased operation
- Rain alert received via Telegram.

5.6 Temperature and Humidity Sensor (DHT11) Testing

Purpose: To validate environmental monitoring.

Test Procedure

- DHT11 sensor data were taken under different conditions of temperature and humidity.
- Values were shown using LCD and Telegram.

Expected Outcome

- Accurate and continuous temperature and humidity readings.

Actual Result

- Sensor data was updated correctly without any delay

5.7 LDR Sensor and Automatic Lighting Testing

Purpose: To validate lighting control based on ambient light.

Test Procedure

- The LDR sensor was subjected to both bright and dark conditions.
- Relay operating for lighting is observed.

Expected Outcome

- Darkness → Light ON
- Bright light → Light OFF

Actual Result

- Lighting system worked automatically as expected.

5.8 IR Sensor and Animal Detection Testing

Purpose: To safeguard crops against animal entry.

Test Procedure

- There was simulation of movement in front of IR sensors.
- Buzzer, fence, and alert behavior were observed.

Expected Outcome

- Animal detection → Buzzer and fence ON with alert.

Actual Result

- Buzzer and fence were triggered properly.
- Alert received via Telegram.

5.9 LCD Display Testing

Purpose: To local monitoring validation via LCD.

Test Procedure

- Sensor values and system messages were observed on the LCD.

Actual Outcome

- The values of the sensors and messages displayed on the system were observed on the LCD display.

5.10 Integrated System Validation

Purpose: To evaluate the performance of the entire system under real-time processing.

Test Procedure

- All the sensors and actuators worked at the same time.
- System response and stability.

Actual Result

- Smooth sensor, actuator, and communication coordination and IoT communication.
- There have been no functional conflicts and any possible delay

5.11 Validation Metrics

sensor / Parameter	Threshold Value	Condition	System Action
Soil Moisture Sensor	<200	Low moisture (dry soil)	Water pump ON
	<300	Sufficient moisture	Water pump OFF
Rain Sensor	<1000	Rain detected	Water pump OFF, alert sent
LDR (Light Sensor)	<1000	Darkness detected	Light ON
	<3000	Bright light	Light OFF
IR Sensor	low	Animal detected	Buzzer & fence ON, alert sent
Temperature (DHT11)	----	Continuous monitoring	Display & IoT update
Humidity (DHT11)	----	Continuous monitoring	Display & IoT update
WI-FI/Telegram	1 second delay	Message polling interval	Status update / alert

Table 5.11 : Validation Metrics

CHAPTER 6

RESULTS AND DISCUSSION

6.1 Results

The IoT Solar Powered Inverter and Smart Irrigation System was fully tested by combining the ESP32 microprocessor with an array of environmental sensors including; temperature and humidity soil moisture sensor, DHT11 sensor rain sensor, LDR, IR animal detection module, relay, buzzer, and electric fence control. Also the Telegram Bot allowed the users to view the real time data as well as allow them to access the system remotely over the internet.

During testing the system operated as expected and the results are outlined below.

1. Successful Wi-Fi and IoT Connectivity:

ESP32 successfully established connection to the “Irrigation123” Wi-Fi hotspot in a couple of seconds. As soon as ESP32 establishes Wi-Fi connection, it creates a secure HTTPS session with Telegram using the Universal Telegram Bot library. As soon as the connection is created with Telegram, a welcome message — “Solar Based Smart Irrigation System...” — is sent to the user which confirms that the IoT has been initiated properly.

ESP32 responds to all Telegram commands correctly as follows:

/hello -> Responds with: “Hello! ESP32 is online.”

/status -> Sends current values of: Soil Moisture, Rain Level, Temperature & Humidity, LDR intensity

Above results show that there is very little delay (~1 sec polling interval) while sending data from ESP32 to Telegram for remote monitoring.

2. Temperature and Humidity Data Collection (DHT11):

Every cycle the DHT11 sensor provides temperature (in °C) and humidity (as %) values. Values obtained from DHT11 are displayed on the 16×2 LCD and are also printed in the serial monitor.

Performance Observed

Temperature readings have $\pm 1^\circ\text{C}$ variation.

Humidity readings vary based upon the moisture present in the environment.

Due to optimal use of delay the LCD displays values continuously without flickering.

Real time display of data on LCD allows for on-site monitoring and data from Telegram will aid in making decisions remotely.

3. Soil Moisture System Functionality:

Soil moisture values were read in by ADC pin (36). Then inverted to scale using (4095 - value).

Actions of the System

moisture < 200

"Low Moisture" on LCD

ON Relay1 → ON Pump

Telegram alert: "Low Moisture Pump On..."

moisture > 300

Pump automatically OFF

Result: Pump started working at correct conditions of low-moisture and stopped once soil is moist enough.

Automatic irrigation eliminates: Manual labor dependency, Water wastage, Over-irrigation, Telegram Alert feature gives the user full awareness of the status of the field even if they are not physically at the field.

4. Rain Detection System Results:

The rain sensor was being continuously polled with ADC pin 39.

Behavior of System

If Rain > 1000 → Rain detected

"Rain Detected" on display

Automatically shuts off Pump

Notification via telegram to user Smart Irrigation needs this, to prevent wasting water when raining.

Results of Testing: Quick & Reliable detection of rain, Instantaneous shut down of pump Good performance of sensors even at different levels of outdoor illumination.

5. LDR-Based Lighting System Performance:

The LDR sensor controls the lighting automatically as a function of ambient light.

Action of System

$LDR < 1000 \rightarrow$ Darkness detected

ON Relay2 $\rightarrow$ ON Field Light

"Darkness Detected" on Telegram

$LDR > 2000 \rightarrow$ sufficient light

OFF Relay2

Results of Testing: Lighting system performed correctly throughout evening and nighttime testing. Enhances visibility of the field and will support irrigation at night and/or security activities.

6. IR Sensor-Based Animal Detection Results:

IR obstacle sensor (pin 4) senses movement near the crop field.

Response of System

When IR signal triggers:

LCD: "Animal Detected"

ON Buzzer

Briefly turn ON Electric Fence

Send Telegram message to user

Reset the system for the next trigger

Results of Testing: Alert and protective response are very reliable and provides protection for crops from animals at night or early morning.

7. Solar Inverter & Power System Performance:

A solar panel-based power supply was utilized in this project as an alternative power source to provide electricity to the ESP32 board, sensors, pump, relay other safety-related components.

Observation

Solar generated enough power to recharge the battery and maintain continuous operation of the entire system. The low power consumption of the ESP32 enabled the system to maintain extended backup times.

6.2 DISCUSSION

This project has integrated IoT, automation, sensor-intelligence, and solar-generated power into a single unified smart-irrigation model. The results indicate that both the software (program logic) and hardware functions operate seamlessly and work well together.

Achievements:

- Fully automated irrigation through real-time soil moisture and rain measurements.
- Environmental monitoring (temperature, humidity, light) is achieved in real-time.
- IR-based animal detection with buzzer and fence enhances crop protection.
- Remotely accessible data and alert capabilities via Telegram messaging app.
- Energy-efficient solar power operation.

System Reliability: The constant loop monitoring, timely updates to LCD, and quick Telegram responses clearly show that the ESP32 program has a good level of reliability. Triggered events (Pump On/Off, Lights, Fence) were implemented promptly.

Performance

The system performed all of the following tasks:

- Reduced labor for irrigation.
- Increased safety within the field.
- Enabled remote monitoring of the farm.

No electrical "noise" was introduced when switching loads with relays; and the microcontroller was never reset by the relay switching. Due to the utilization of solar-generated energy, this system is very applicable for use in rural and outdoor agricultural settings.

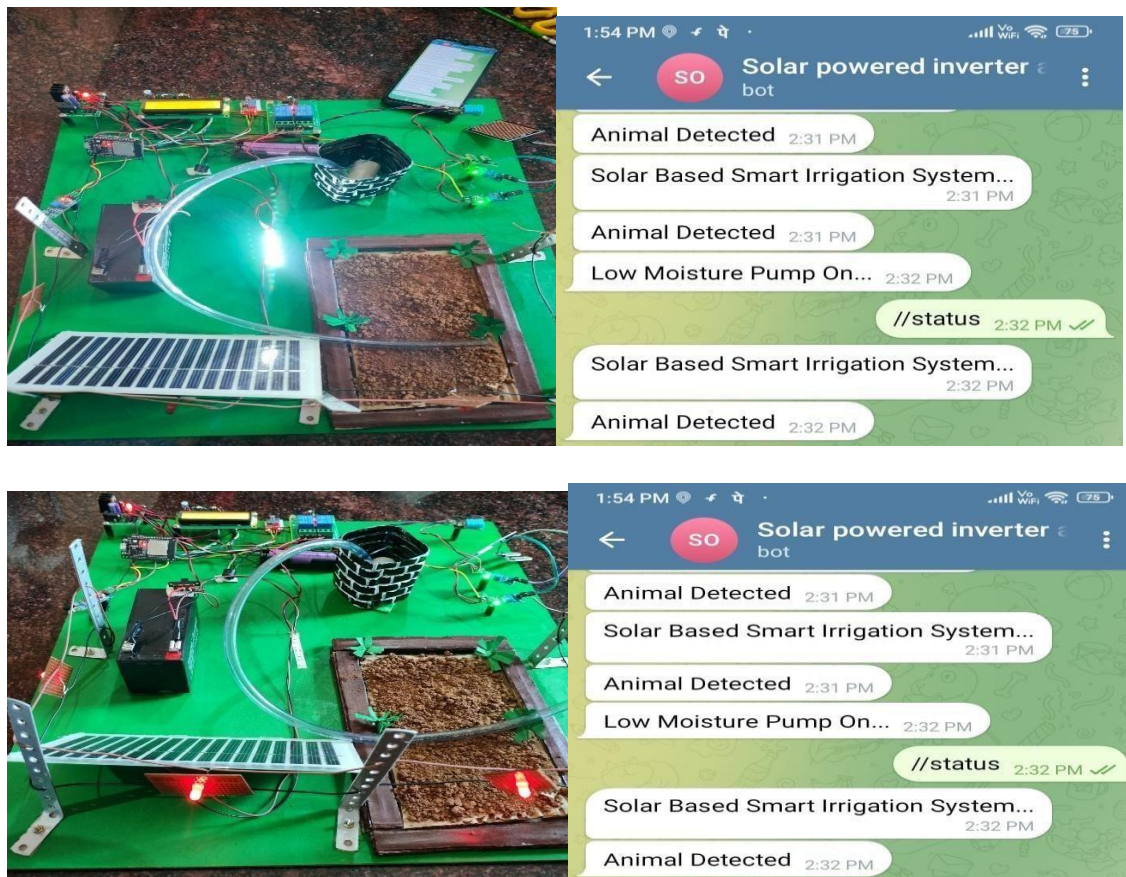


Fig 6.3 Results of IoT-Based Solar Powered Inverter and Smart Irrigation System

CHAPTER 7:

APPLICATION S AND ADVANTAGES

7.1 APPLICATIONS

1. Intelligent Agriculture & Precision Farming

One of the applications of the proposed system is precision farming or smart farming. With the proposed system, the farmer will have the ability to irrigate his or her crops according to the soil moisture, temperatures, and humidity levels. This ensures that the crops will not be irrigated when not really needed and will therefore grow to the highest potential. Precise farming will therefore assist in boosting the yield of crops as well as cutting costs.

2. Water Conservation in Drought-Prone Areas

Water scarcity is a problem in many parts of the world. This smart irrigation system is beneficial in the sense that it prevents the water from being over-irrigated and pumped. The rain sensor feature further enhances the concept of saving water by not irrigating the crops when it is raining. This smart irrigation system is very useful in areas that suffer from droughts.

3. Greenhouses and Polyhouse Farming

This system can be effectively employed in greenhouses as well as polyhouses, where a controlled environment is essential. This system allows the measurement of temperature, humidity, and soil moisture values continuously. Thus, based on these values, irrigation as well as climatic conditions can be controlled accurately. This is because the mentioned system is fully automatic.

4. Intelligent Farming with IoT-Based Remote Monitoring

Internet of Things (IoT) integration helps farmers remotely monitor their fields with the assistance of a mobile device. Notifications regarding real-time soil moisture, rainfall, light, and animal entry alerts will help in taking immediate actions regarding the status of the farmland. These are very useful

5. Horticulture and Plantation Crops

This requires uniform and correct irrigation in horticulture crops, including fruits, flowers, and plantation crops. Infrared technology-based irrigation presented in the project ensures equal watering of plants based on soil moisture content, and it also promotes uniform growth in plants. This, in turn, reduces stress on crops and thereby improves their quality.

6. Intelligent Gardening and Landscaping

The system can additionally be applied in private gardens, parks, and for landscaping. The plants are able to get enough water using automated irrigation, without any effort at all. The solar power system is additionally very cost-effective and environmentally friendly for such gardens.

7. Sustainable and Eco-Friendly Farming Practice

It helps in sustainable farming by encouraging the use of resources in the right manner. The carbon emissions are decreased by using solar energy, which is renewable, whereas the optimization of water use will ensure conservation of natural resources. It would also be apt for farming using eco-friendly and organic methods.

8. Research and Educational Institutions

This proposed system will find its application in agricultural research centers and institutes of learning where such a model will be required. This will be of great use to the students and researchers to study IoT-based automated systems, integration of renewable energy sources, and methods of smart irrigation.

7.2 ADVANTAGES

1. Water Resource Management

One of the most significant advantages of the new system is that it is capable of controlling the water resource effectively. This is achieved because the system is able to detect the soil moisture and release the water as the need arises. In this way, water is not wasted through excessive irrigation.

2. Use of Renewable Energy Resources

This is a solar power-based system, and it is backed by an inverter, hence making the system environmentally friendly. This means that the system depends less on the conventional source of power or fossil fuel. This is a solar power-based system, hence ideal for sustainable agriculture and reduction of carbon emissions.

3. Applicable for Remote & Rural Areas

Agricultural lands lie in remote areas. Power supply cannot be accessed in these areas. However, by the application of this solar inverter, agricultural development can take place in remote areas. This means the solar inverter can also work well in rural and semi-urban areas.

4. Automation & Minimized Human Interaction

The smart irrigation system is also able to automate all the major operations involved in farming, including the pumping of water, supply of light, and guarding of the crops. This is one of the major merits of the smart irrigation system since the operations require less supervision by the farmer.

5. Real-Time Monitoring Through IoT Technology

Through IoT integration, real-time monitoring and notification systems can be enabled through the application of a mobile application. Through a mobile application, the farmer is capable of receiving real-time notifications pertaining to soil moisture, temperature, humidity, rainfall, light, and the presence of cattle.

6. Scalability and Flexibility

It is easy to expand by adding more sensors and irrigation zones. It is also easy to customize by changing the threshold values or control statements. This is done using software.

7. Prevention of Over-Irrigation

The presence of a rain sensor ensures that watering is discontinued when it rains. This goes a long way in avoiding watering and ensuring that crops are not waterlogged. The control logic ensures that watering takes place optimally based on the environment created.

8. Energy Efficiency

The system controls the electrical devices, including water pumps and lightings, only when the need arises. This smart control enables the reduction of energy consumption and the subsequent optimization of energy efficiency. The use of solar energy and the automated system results in the reduction of costs incurred in the long run.

9. Better Crop Protection

The IR sensor-based animal detection feature is useful in crop protection against damage by animals. In case of intrusion, it triggers an alert, and an electric fence and buzzer are switched on.

10. Automated Lighting Systems

The lighting system is designed to automatically turn ON during low light conditions and turn OFF during adequate daylight. This helps to ensure proper visibility during night operations.

11. User-Friendly

The LCD display allows sensor data and system conditions to be viewed locally, making it easy to understand and work with. With mobile-based IoT monitoring, there is also remote accessibility.

CHAPTER 8:

CONCLUSION & FUTURE SCOPE

8.1 Conclusion

The rise in demand for energy, water resources, and sustainable agriculture practices has triggered the design and development of intelligent systems, which involve the integration of renewable energy sources with automation technology. The proposed Internet of Things (IoT) Based Solar Powered Inverter and Smart Irrigation System assists in overcoming the above-said limitations by integrating solar power generation, management, and smart irrigation through the Internet of Things. The proposed model effectively uses solar power as a major source to power the inverter, using which the entire agriculture load and water are pumped, ensuring a continuous power supply. The model uses the capability of IoT-based sensors like Soil Moisture Sensors, Voltage and Current Sensors, and Environmental and Monitoring Units, which continuously monitors the data from the agriculture field. The data obtained from the agriculture fields is transmitted into a Cloud Platform, wherein the data can be accessed and analyzed with a web or mobile interface. The Smart Irrigation System enhances the automation of water supply based on soil moisture levels, eliminating over-irrigation and under-irrigation. This ensures water conservation and maximized crop production. The automation also reduces human intervention and assures integral water supply even at remote or difficult-to-reach agricultural fields. The model also proves highly effective for farmers in the rural sector, wherein electricity supply is fallible, and farmers are forced into manual water irrigation, which is inefficient and time-wasting. The solar-powered inverter increases the reliability of the model by supplying pure and green renewable energy source independent of main electricity and non-renewable energy sources. Battery Backup ensures adequate supply at a time of restricted solar power generation, and the monitoring aspects pertaining to IoT helps in tracking the monitoring and energy usages of the model. Integration of Alert and Status features enhances the accuracy and maintenance of the model. Implementation-wise, the model is a significant one and has effectively shown the workability of concepts such as integration of sensors, data acquisition, Cloud connectivity, and control from anywhere in the world. The model has a modular and economical design and could effectively work in real-time agriculture and power for underdeveloped farming and power sector. The case study results have effectively validated the efficiency, reliability, and green environment provided by the proposed model. Consultatively, it could be said that the IoT-Based Solar Powered Inverter and Smart

Irrigation System is a green and efficient model for agriculture and water conservation. It helps to develop smart farming projects, supports the adoption of renewable energy sources, and promotes efficient use of resources. The project achieves its objectives and provides an excellent foundation for future innovation in the areas of smart agriculture and energy resources.

8.2 Future Scope

1. Artificial Intelligence and Machine Learning

It consists of processes like integration of Artificial Intelligence and Machine Learning algorithms. The purpose of integration is to generate predictions for irrigation with the help of Machine Learning algorithms, with the previous information regarding soil moisture content, environmental conditions, type of crop, and irrigation timing applied to it. The irrigation timing helps to reduce water consumption as well as optimize crop growth at a time. The AI algorithm has the ability to access irregular information represented by the sensor in data.

2. Integration of Weather Forecast and Climate Data

Incorporating the forecasts of the weather into this system will be a great improvement in the decisions concerning irrigation. The probability of rainfall, temperature, humidity, and even the speeds of the wind can be downloaded from the online weather APIs. The irrigation timing will be automatically postponed in case of rain since the system will be able to decide on the timing of irrigation and this timing will be postponed if it is rainy weather. Climate data will enable the farmer to decide on planting.

3. Advanced Energy Management System

The solar inverter is still left for improvement regarding the use of the intelligent energy management system. This will have the ability to divide the priority use of the energy in the form of the irrigation, domestic, and other bills. The analysis of the power will also enable the understanding of the power use and hence increase efficiency. The system will also have the ability to charge the batteries.

4. Battery Health Monitoring and Predictive Maintenance

Future improvements that could be made for this system These are the features which could be implemented for this system in the future: Battery health: Battery attributes like voltage,

current, temperature, or the number of charge cycles could be tracked. Notification for maintenance and replacement based on data analysis could be performed. This could extend the life of this system.

5. Improve Mobile Application and User Interface

A customized mobile application developed for both android and ios operating systems would enhance the interface. In this respect, the mobile-based system would be assisting in remote monitoring of solar power, conditions of the inverter, battery health, as well as the drip irrigation system. The possibility of sending alert notifications/messages to the user after any default/error condition has occurred or reduced battery levels or sensor anomalies cannot be ruled out. The multilingual mobile application makes it more agriculture-friendly.

6. Multi-Zone and Large Scale Irrigation

The irrigation zone control system can be developed. The independent soil and crop condition control corresponding to each zone can also be independently developed. Its advantage is that it makes the invention more applicable on large agricultural land. The zone control irrigation watering system helps in providing maximum quantity with no wastage of water. The inverter capacity used is also upgradeable for heavy load processing.

7. Hybrid Power and Grid Synchronization

Future Improvement There should be the addition of operation for hybrid power with synchronization between the solar inverter and the grid or other forms of alternative power. This should be able to switch automatically between solar power, batteries, and power from the grid based on level connections. Net metering should be included in this device.

8. Additional Sensor Integration

An angle that is not being utilized The proposed system can further be improved with the incorporation of other sensors like temperature sensors, humidity sensors, pH sensors, nutrient sensors, and light intensity sensors. The above-mentioned sensors provide all the necessary information needed in precision farming. Optimum fertilizer application is determined with the help of pH values and nutrient levels in the soil, whereas temperature and humidity sensors determine crops.

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